Project Proposal: *Boon Sop: The Unbroken Debt*

3D Atmospheric Web-Based Horror Game

Executive Summary

Boon Sop: The Unbroken Debt (▓▓▓▓▓▓▓▓▓▓ ▓▓▓▓▓▓▓▓▓▓▓▓▓▓) is a 3D first-person atmospheric survival horror game engineered specifically for web browsers via WebGPU / WebGL2 (powered by Babylon.js or PlayCanvas). Set in 1932 French-Colonial Cambodia, the project combines rich Khmer folklore, historical mystery, and high-tension survival horror.

Players take on the role of Sora, a 24-year-old graphic designer from Phnom Penh who returns to her rural roots after a alarming phone call to break an ancestral pact made with a land guardian spirit (*Neak Ta*). Structured as a 5-episode seasonal series, the game offers 15–20 hours of gameplay without requiring external downloads or high-end gaming hardware.

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│                        PROJECT AT A GLANCE                                   │
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- Platform: Desktop Web Browsers (Chrome, Firefox, Safari, Edge)
- Technology: Babylon.js / PlayCanvas (WebGPU / WebGL2)
- Genre: 3D First-Person Atmospheric Survival Horror
- Format: 5 Episodic Chapters (3–4 Hours per Episode)
- Target Audience: Fans of psychological horror, narrative indie games, and cultural folklore (e.g., Silent Hill, Fatal Frame)

Project Vision & Core Value Proposition

1. Cultural Authenticity & Unique Setting

While modern horror games frequently draw from Western or East Asian folklore, Boon Sop taps into 1930s Cambodian heritage—incorporating traditional architecture, classical music instruments (*Pinpeat*), spiritual rituals (*Sai Sin*, *Bay Ben*), and supernatural entities (*Neak Ta*, *Pret*, *Khmaoch Prey*).

2. Frictionless Accessibility

By leveraging modern web technology (WebGPU / WebGL2), players can launch high-quality 3D interactive experiences directly through a browser link, eliminating long download times, installer friction, and strict platform restrictions.

3. Deeply Immersive Diegetic Mechanics

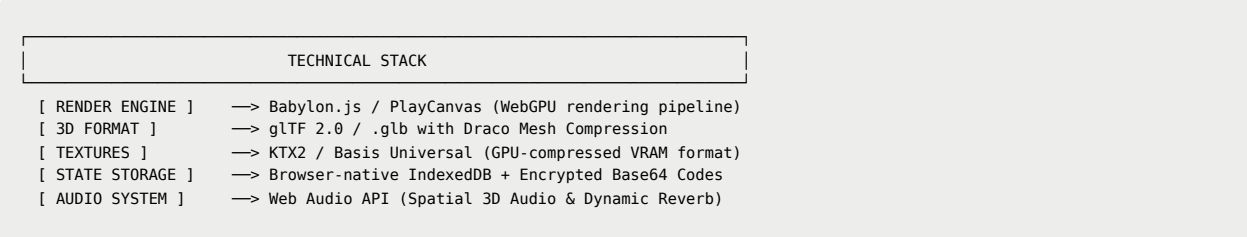
All user interfaces—maps, lore notes, light levels, inventory, and health—exist entirely within the 3D world space (via Sora's physical leather satchel and sketchbook), driving immersion.

Episode Roadmap & Narrative Arc

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[ EPISODE 1 ] ──> The House of Mourning (Battambang Stilt House)
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[ EPISODE 2 ] ──> The Whispering Pagoda (Pchum Ben & Stupa Courtyard)
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[ EPISODE 3 ] ──> The Drowned Village (Tonle Sap Flooded Clinic)
                  |
[ EPISODE 4 ] ──> The Rubber Plantation (Kampong Cham Colonial Estate & Fog Grid)
                  |
[ EPISODE 5 ] ──> The Sunken Sanctuary (Phnom Kulen Mountain Catacombs)
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Episode	Setting	Core Gameplay Mechanic	Seal Artifact
Ep. 1: The House of Mourning	Battambang Teak Stilt House	Resource Management (Incense & Kerosene)	Bronze Token
Ep. 2: The Whispering Pagoda	Village Pagoda & Stupa Yard	Audio Stealth & Line-of-Sight (Pret)	Silver Betel Box
Ep. 3: The Drowned Village	Flooded Tonle Sap Clinic	Boat Navigation & Moisture/Dryness	Terracotta Water Jar
Ep. 4: The Rubber Plantation	Kampong Cham Colonial Estate	Compass Grid & "Voice Mimic" Taboo	Iron Tapping Knife
Ep. 5: The Sunken Sanctuary	Phnom Kulen Stone Catacombs	Elemental Trials & Severance Ritual	Golden Inscription Slab

Technical Stack & Architecture



Performance & Target Benchmarks

- Target FPS: 60 FPS @ 1080p (Desktop Discrete GPU) / 30 FPS @ 720p (Integrated Laptop GPU).
- Draw Call Budget:

≤ 120

- draw calls per frame using static mesh instancing.
- Polygon Limit:

$\le 150,000$

- visible polygons per frame with Level-of-Detail (LOD) distance switching.Package Budget:

$\le 85\text{ MB}$

- initial asset load per episode.

Production Schedule & Milestones

The development cycle is projected over an 11-month timeline, divided into four major phases:



Phase 1: Pre-Production & Engine Setup (Months 1–2)

- Finalize Technical GDD, 3D asset list, and audio specs.
- Establish Babylon.js / PlayCanvas WebGPU rendering environment.
- Build player movement controller, camera head-bob, and raycast interaction systems.

Phase 2: Core Mechanics & Episode 1 Vertical Slice (Months 3–5)

- Implement diegetic 3D satchel/sketchbook UI and lighting engine (kerosene lamp + matchbox).

- Develop base Stealth AI perception model (Visual Cones + Spatial Audio Impulse).
- Build and release Episode 1 (*The House of Mourning*) as a playable internal vertical slice.

Phase 3: Content Production & Episode Rollout (Months 6–8)

- Construct environment modules for Episodes 2 through 4.
- Implement special AI variants (*Pret* high-hearing logic, *Chhlong Tonle* water detection, *Khmaoch Prey* voice-mimic system).
- Integrate cross-episode save state persistence using IndexedDB.

Phase 4: Episode 5 Climax, QA, & Public Launch (Months 9–11)

- Build Episode 5's trial chambers, boss ritual logic, and multiple ending sequences.
- WebGL/WebGPU cross-browser compatibility testing (Chrome, Safari, Firefox, Edge).
- Texture compression tuning (KTX2) and memory profiling to eliminate tab crashes.
- Global web release and launch marketing campaign.

Resource Requirements & Team Structure

Role	Primary Responsibilities
Lead Web Game Developer	Engine setup (Babylon.js/PlayCanvas), shader programming, WebGPU optimization.
Technical Designer / Scripter	Gameplay state machines, puzzle mechanics, stealth AI logic implementation.
3D Environment & Prop Artist	Period-accurate 1930s Khmer architecture, props, and Draco/glTF optimization.
Sound Designer & Audio Engineer	Spatial 3D soundscapes, traditional Pinpeat ambient scoring, and horror sound cues.
Writer & Cultural Consultant	Dialogue scripts, palm-leaf manuscript translations, historical folklore accuracy.

Risk Assessment & Mitigation Strategies

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│                     RISK MANAGEMENT MATRIX                           │
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  [ RISK 1: Web Memory Leaks & Browser Crashes ]
  ──> MITIGATION: Strict asset loading/unloading between scenes; KTX2 texture
      compression; continuous VRAM profiling during QA.

  [ RISK 2: High Latency / Slow Web Page Load ]
  ──> MITIGATION: Lazy loading assets per room; asynchronous audio streaming;
      keeping total initial load under 85 MB.

  [ RISK 3: Cultural Inaccuracy or Misrepresentation ]
  ──> MITIGATION: Collaboration with Cambodian cultural advisors and elders on
      traditional practices, language, and spiritual customs.
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Conclusion & Next Steps

Boon Sop: The Unbroken Debt

bridges cultural folklore with modern browser technology, delivering a web-native survival horror experience.